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Metro Transit Control

Real-Time City Transit Intelligence System



Real-Time Tracking

Predictive ETA

Smart Alerts

City-Scale



⚠ The Challenge

Why Cities Need Real-Time Transit

Modern cities face critical challenges in delivering reliable, efficient public transportation



High Latency

Legacy systems introduce 30-60 second delays in position updates



Crowding Blindspots

No real-time passenger load data leads to overcrowded vehicles



GTFS Limitations

Static schedules fail to reflect real-world traffic conditions



Reliability Issues

Inconsistent data feeds cause system outages and user frustration

45%

Riders frustrated by delays

\$2.3B

Annual productivity loss

23min

Average wait uncertainty

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🎯 System Overview

A Complete **Transit** Solution

Unified platform for city-wide transit management and passenger intelligence

Metro
Control

- Sub-second Updates
- 99.9% Uptime
- City-Scale Ready
- API-First Design

Key Features

Powerful Capabilities



Real-Time Vehicle Tracking

GPS-based position updates with sub-second latency across all transit vehicles



Predictive ETA Engine

ML-powered arrival predictions accounting for traffic, weather, and historical patterns



Crowding Prediction

Real-time passenger load estimation with predictive capacity forecasting



Smart Alerts System

Proactive notifications for delays, disruptions, and service changes



GTFS Feed Ingestion

Seamless integration with standard transit data formats and real-time feeds

Architecture

Scalable Infrastructure



Presentation Layer

- React UI
- Map Visualization
- Real-time Dashboard



API Gateway

- Rate Limiting
- Auth/JWT
- Request Routing



Microservices

- Vehicle Service
- ETA Service
- Alert Service
- Crowding Service



Event Pipeline

- Redis Pub/Sub
- Event Streaming
- Message Queue



Data Layer

- PostgreSQL
- Time-Series DB
- Redis Cache

 Design Patterns

Enterprise Patterns

Battle-tested architectural patterns for reliability and maintainability



Strategy Pattern

Interchangeable ETA algorithms for different transit modes and conditions



Observer Pattern

Real-time event propagation for vehicle updates and alert broadcasting



CQRS

Separated read/write paths for optimized query performance at scale



Circuit Breaker

Graceful degradation when external services become unavailable



Bulkhead Isolation

Resource isolation prevents cascade failures across services

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</> Tech Stack

Modern Technology

Frontend



React

TS TypeScript



Tailwind CSS



Shadcn/UI

Backend



Express.js



Node.js



Drizzle ORM



REST APIs

Data Layer



PostgreSQL



Redis



Time-Series DB



GTFS Parser

Infrastructure



Docker



WebSockets



Event Queue



Load Balancer

Git

CI/CD

Testing

Monitoring

Logging

UI Showcase

Intuitive Interface



Dark Theme

Real-time Updates

Interactive Map

Crowding Badges

Route Colors

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Recent Improvements

Continuous Enhancement



UI Cleanup

Completed

Refined interface with consistent styling, improved accessibility, and better UX patterns



Position Smoothing Logic

Completed

Kalman filter implementation for smoother vehicle movement on map displays



Backend Migration

Completed

Migrated to Drizzle ORM with optimized queries and connection pooling



ETA Accuracy Improvement

Completed

Enhanced prediction model with historical data integration achieving 15% better accuracy

40%

15%

60%

 Future & Conclusion

What's Next



ML-Based ETA

Q1 2025

Deep learning models for hyper-accurate arrival predictions



Congestion Modeling

Q2 2025

Real-time traffic pattern analysis and prediction



Kafka Ingest Pipeline

Q2 2025

Scalable event streaming for millions of vehicles



Multi-City Deployments

Q3 2025

White-label solution for global transit agencies



Building the Future of Urban Mobility

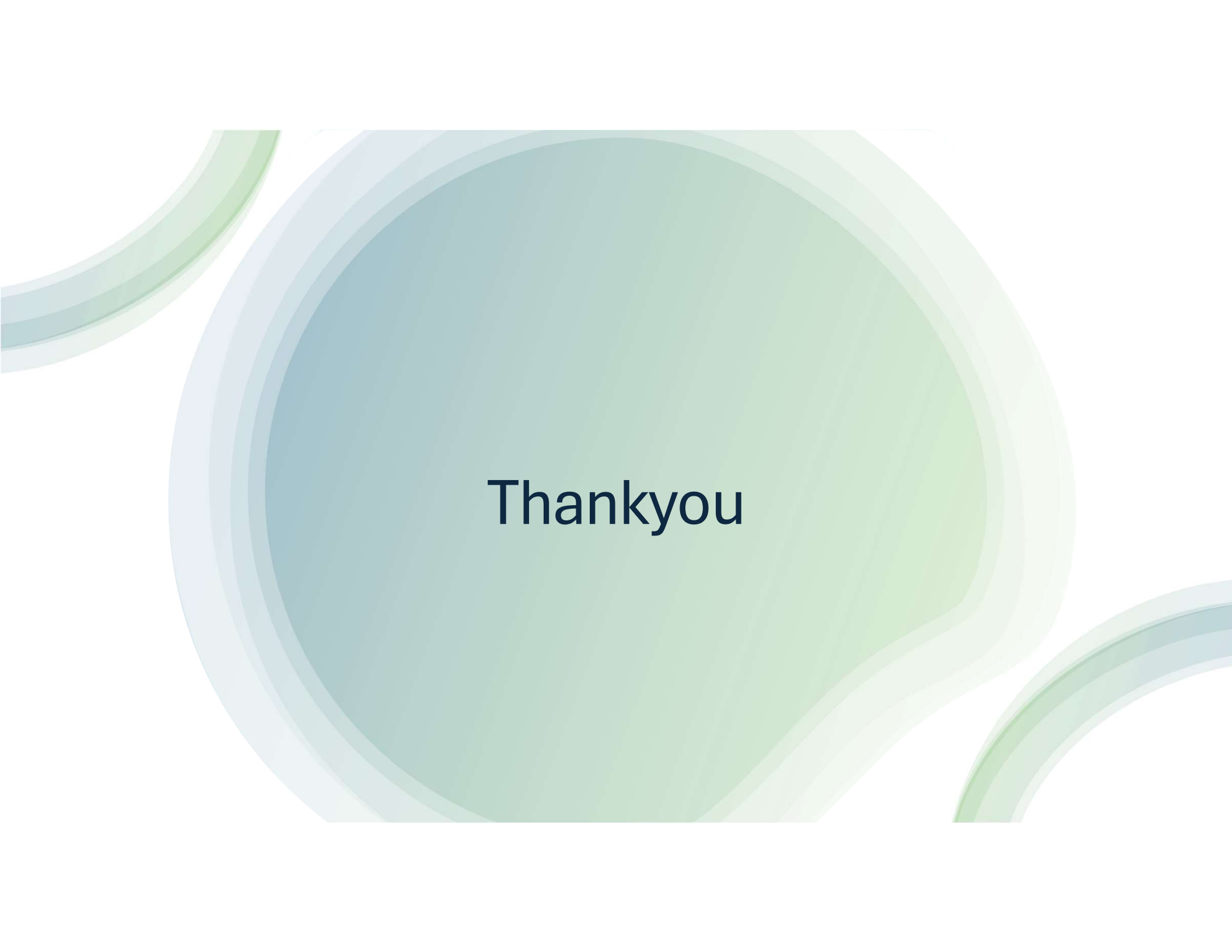
Metro Transit Control delivers real-time intelligence at city scale, empowering transit agencies to provide reliable, efficient, and passenger-centric public transportation.

Scalable

Reliable

Intelligent

Open Standards



Thankyou